INORGANIC CUME

Feb-20-2009

Last week's seminar was given by Prof. Ken Balkus from UT-Dallas on syntheses, surface modifications, and applications of TiO2-based nanotube. Please try to answer the following questions that a typical graduate student may ask.

(1). Use your sketch to show the edge-sharing and corner-sharing octahedra of [TiO_6] as the two basic crystal construction manners in the nanotube's crystal lattice. (20')

(2). If the nanotube's surface would be approximately the same as a typical TiO2 bulk crystal surface in atomic scale, how would you functionalize the surface so as to make the nanotube surface to be hydrophobic and hydrophilic, respectively? (20')

(3). It is known that TiO2-nanoparticles are the major ingredient in sunscreen lotion. Predict advantage(s) and disadvantage(s) in the development of a new sunscreen using the TiO2-nanotube as the major ingredient. (20')

(4). How would you use the TiO2-nanotubes to study H_2-storage, drug release, and water photolysis? Please use examples or simple sketches to briefly discuss your chemistries in each of the three applications. (30')